Koustav Majumder

Software Engineer

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SUMMARY

Software Engineer with 2.5 years of experience developing Java and Spring Boot microservices, REST APIs, and asynchronous workflows for financial services and enterprise platforms. Experienced across the software development lifecycle, including requirements clarification, feature development, API integration, automated testing, CI/CD, deployment, production troubleshooting, and performance optimization. Skilled in Java 8+, object-oriented programming, design patterns, Spring Data JPA, Hibernate, SQL, relational and NoSQL database concepts, Git, Jenkins, and Agile/Scrum delivery. Supported platforms handling 1M+ monthly API requests, improved API latency and query performance by up to 40%, and strengthened automated test coverage and release reliability.

EXPERIENCE

Community Dreams Foundation (CDF), Chicago IL | Backend Developer — Aug 2026 – Present

- Designed and developed Java and Springboot backend services for Elpsan Bio, a research platform focused on automated phenotype extraction and therapeutic discovery.
- Implemented service-to-service communication and JSON-based REST APIs for high-volume transactional workflows, reducing average API latency by 28%.
- Designed and optimized relational data access using Spring Data JPA and Hibernate with Oracle and PostgreSQL; applied indexing, query tuning, and connection pooling to reduce query execution time by 40%.
- Implemented automated unit and API integration tests using JUnit and WireMock, raising automated test coverage to 85% and reducing regression defects by 40%.
- Improved Jenkins CI/CD pipelines for automated builds, testing, and deployments to PCF, reducing release time by 45% and supporting zero-downtime deployments.
- Collaborated with cross-functional teammates to clarify feature requirements, discuss API and service design decisions, and align implementation and testing with expected behavior. Maintained application stability, validated changes, and supported reliable releases.

Capgemini Technology Services, India | Software Engineer — Aug 2021 - Sep 2023

- Developed scalable Java and Spring Boot REST APIs for Discover Financial Services' digital banking platform, reducing response latency by 40% through asynchronous processing.
- Contributed to a microservices architecture with emphasis on security, compliance, and high availability.
- Built and maintained Jenkins CI/CD pipelines to streamline build, test, and deployment workflows, enabling zero-downtime releases across five PCF/OpenShift environments.
- Analyzed production logs and application metrics to help sustain 99.9% platform uptime and reduce incident resolution time.
- Established automated unit and integration testing with JUnit and WireMock to improve code coverage and deployment reliability.
- Worked with cross-functional teammates to clarify assigned feature requirements, evaluate implementation approaches, and communicate API/service design decisions. Took ownership of engineering tasks through development, testing, and release.

Annexbyte IT Solutions, India | Software Developer Intern — Mar 2020 - Jun 2020

- Built REST endpoints and service-layer features in an existing Spring Boot application, taking assigned features from development through testing and production release in two-week sprints.
- Optimized MySQL data access by replacing N+1 query patterns and unindexed lookups with tuned JPA queries and stored procedures, reducing order-history page-load time from approximately six seconds to under two seconds.
- Created Dockerfiles and OpenShift deployment configurations for two Spring services, reducing environment provisioning time from two days to a few hours.
- Added JUnit and Mockito tests for legacy service classes and resolved defects identified during QA regression cycles.
- Worked with the team to maintain existing code, improve service reliability, and deliver updates within Agile/Scrum workflows.

EDUCATION

Master of Science in Computer Science | Illinois Institute of Technology, Chicago — **May 2026**

Bachelor of Technology in Computer Science Engineering | Amity University, Kolkata, India — **May 2021**

SKILLS

Languages: Java (8+), Core Java, SQL, PL/SQL, JavaScript

Frameworks & APIs: Spring Boot, Spring Framework, Spring Data JPA, Hibernate, RESTful APIs, JSON

Architecture & Integration: Microservices Architecture, Object-Oriented Programming (OOP), Design Patterns, Service-to-Service Communication, Asynchronous Processing, REST API Integration

Databases: Oracle, Kafka, PostgreSQL, MySQL, NoSQL Database Concepts, Relational Data Modeling, Indexing, Query Optimization

Testing & Quality: JUnit, Mockito, WireMock, Unit Testing, Integration Testing, Regression Testing, Defect Resolution

Cloud, DevOps & Tools: AWS, Docker, OpenShift (OCP), Pivotal Cloud Foundry (PCF), Jenkins, CI/CD, Git, Maven, Gradle

Methods & Support: Agile/Scrum, SDLC, Production Monitoring, Log Analysis, Troubleshooting, Cross-functional Collaboration

PROJECTS

AI-Powered Email Writer & Reply Assistant

- Built a Gmail Chrome extension with a Spring Boot backend and React frontend to generate context-aware email replies using the Gemini LLM.
- Developed REST endpoints to analyze email context and tone preferences; integrated custom Gmail UI controls using Chrome Extension Manifest V3 and content scripts.

MyBrandbae – A Creator Marketplace (Live)

- Streamlined influencer deal lifecycle end-to-end by building a 2-sided creator marketplace with Instagram analytics integration, multi-creator campaign support, and a 6-stage collab workflow using Node.js, Express, and PostgreSQL.
- Cut creator payment delays by 90%+ from 3–6 months to 48 hours post content delivery by integrating Razorpay with escrow-style payment release, webhook verification, and automated email notifications at each stage.
- Ensured platform reliability across 3 user roles (Brand, Creator, Admin) by writing 87 unit/integration tests across 5 test suites using Jest + Supertest, with JWT + Google OAuth auth, role-based access control, and 10 admin endpoints for creator/brand lifecycle management.